CMP5101 Data Mining

Assignment I

Exploratory Data Analysis on Turkey & KKTC Postal Codes Dataset

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1 Introduction

This report presents an exploratory data analysis (EDA) of the Turkey and Northern Cyprus (KKTC) Postal Codes Dataset. The dataset is publicly available on Kaggle at the following link:

https://www.kaggle.com/datasets/erogluegemen/turkey-postal-codes-dataset-2025/code

The dataset provides a hierarchical mapping of Turkey's administrative units down to the neighbourhood (mahalle) level, along with their associated postal codes and geographic coordinates. It covers all 81 provinces of Turkey plus the 6 districts of Northern Cyprus (KKTC).

The goals of this analysis are to understand the structure and quality of the dataset, identify meaningful patterns and relationships between features, detect anomalies or data quality issues, and extract actionable insights.

2 Dataset Overview

2.1 Loading and Structure

The dataset is stored as a semicolon-delimited CSV file with a comma as the decimal separator for coordinate values. It contains **74,342 records** and **7 columns**.

2.2 Feature Description

Table 1 describes each column, its semantic type, and a brief explanation.

Table 1: Feature description of the dataset

Column	Original Name	Type	Description
`il`	İl	Categorical (nominal)	Province name
`ilce`	İlçe	Categorical (nominal)	District name
`semt`	Semt/Bucak/Belde	Categorical (nominal)	Township or town
`mahalle`	Mahalle	Categorical (nominal)	Neighbourhood or village
`PK`	Posta Kodu	Quasi-categorical	Postal code
`lat`	Latitude	Numeric (continuous)	Geographic latitude
`lon`	Longitude	Numeric (continuous)	Geographic longitude

The four categorical columns form a strict administrative hierarchy: `il` ⊃ `ilce` ⊃ `semt` ⊃ `mahalle`. The postal code `PK` is stored as a string to preserve leading zeros and is best treated as a quasi-categorical identifier. The coordinate columns are the only truly numeric features.

3 Summary Statistics

3.1 Categorical Columns

Table 2 summarises the cardinality (number of unique values) of each categorical column.

Table 2: Unique value counts for categorical columns

Column	Unique Values	Notes
Province (`il`)	81	80 TR provinces + KKTC
District (`ilce`)	898	
Township (`semt`)	2,023	
Neighbourhood (`mah`)	53,339	
Postal code (`PK`)	2,818	

On average, each postal code covers 74,342/2,818 $\approx$ 26 neighbourhoods, confirming that postal codes operate at the township or district level rather than per-neighbourhood.

3.2 Numeric Columns

Table 3 shows summary statistics for the coordinate columns, computed on Turkey (TR) records only since KKTC records have no coordinates.

Table 3: Summary statistics for Latitude and Longitude (TR records only, $n = 74{,}071$)

	Mean	Std	Min	25%	75%	Max
Latitude	39.63	1.52	35.90	38.40	40.94	42.03
Longitude	35.40	4.67	25.91	31.74	39.28	44.57

The latitude range (35.9°N – 42.0°N) and longitude range (25.9°E – 44.6°E) are fully consistent with Turkey's geographic bounding box, indicating no out-of-bounds coordinate anomalies.

4 Missing Data Analysis

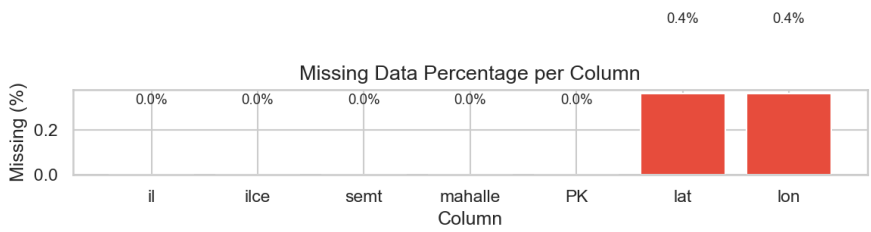


Figure 1: Percentage of missing values per column

As shown in Figure 1, only the `lat` and `lon` columns contain missing values, each at **0.36%** (271 records). All 271 missing coordinate records correspond exactly to the KKTC entries. Turkey's records have complete coordinate coverage.

No empty-string values were found in any of the four categorical columns.

5 Duplicate Detection

The dataset contains **1,189 fully duplicated rows** (identical across all 7 columns). An additional analysis revealed that these duplicates are not scattered randomly across provinces.

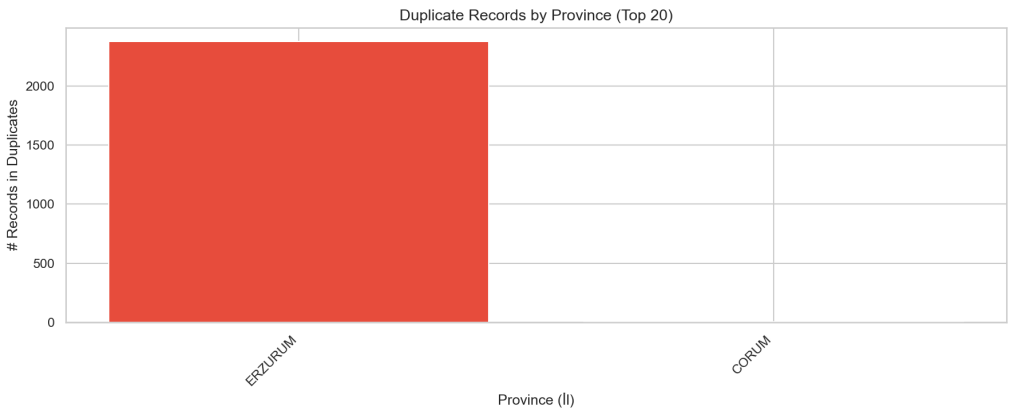


Figure 2: Duplicate records by province

As Figure 2 shows, **2,376 out of 2,378 records involved in duplicates belong to Erzurum province**. This strongly suggests that the entire Erzurum province was entered twice during data compilation, rather than random data entry errors. Only 2 duplicate records belong to another province (Çorum). The duplicates are exact clones — no coordinate discrepancies exist between copies.

6 Distributions: Histograms and Box Plots

6.1 Postal Code Distribution

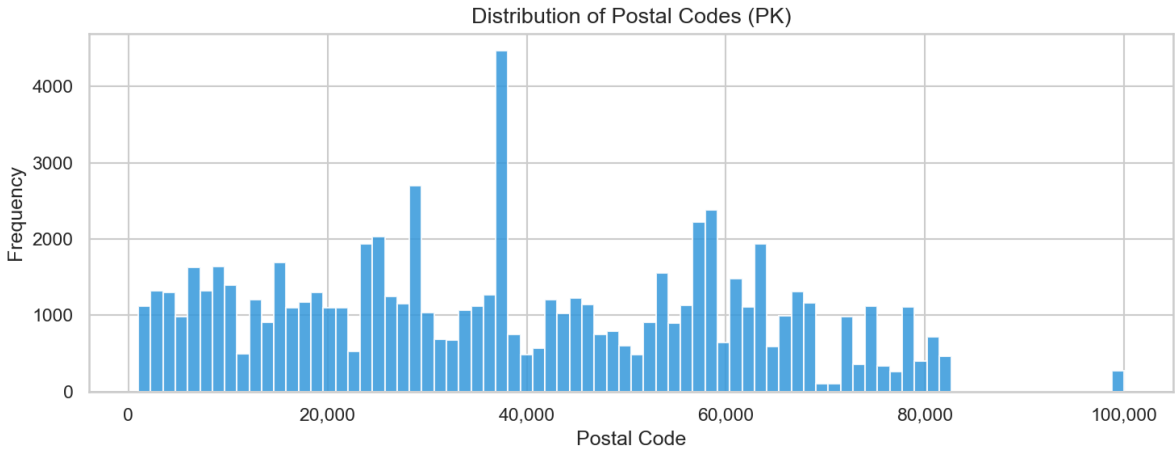
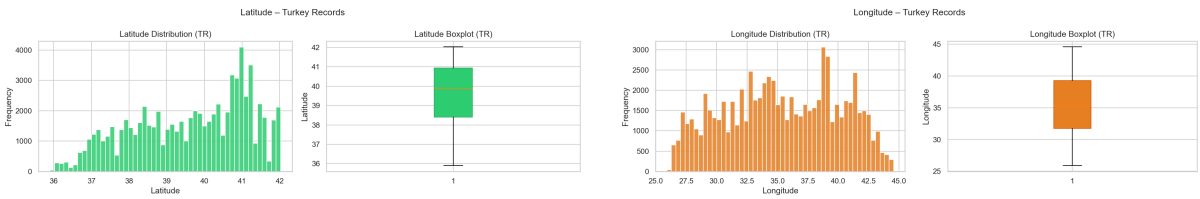


Figure 3: Distribution of postal code values

Figure 3 shows that postal codes are distributed across the full range 01010–81999 for Turkey and 99000–99999 for KKTC. The distribution is roughly uniform across the Turkish range, with a visible gap around the 82000–99000 range (unassigned codes), and a small cluster at the high end representing KKTC.

6.2 Latitude and Longitude



(a) Latitude distribution (b) Longitude distribution

Figure 4: Latitude and longitude histograms and box plots (TR only)

Both distributions are roughly unimodal and symmetric. Latitude is centred around 39.6°N (central Anatolia) with a standard deviation of 1.5°. Longitude is centred around 35.4°E with a wider spread (std 4.7°), reflecting Turkey's elongated east–west shape.

6.3 Province and District Record Counts

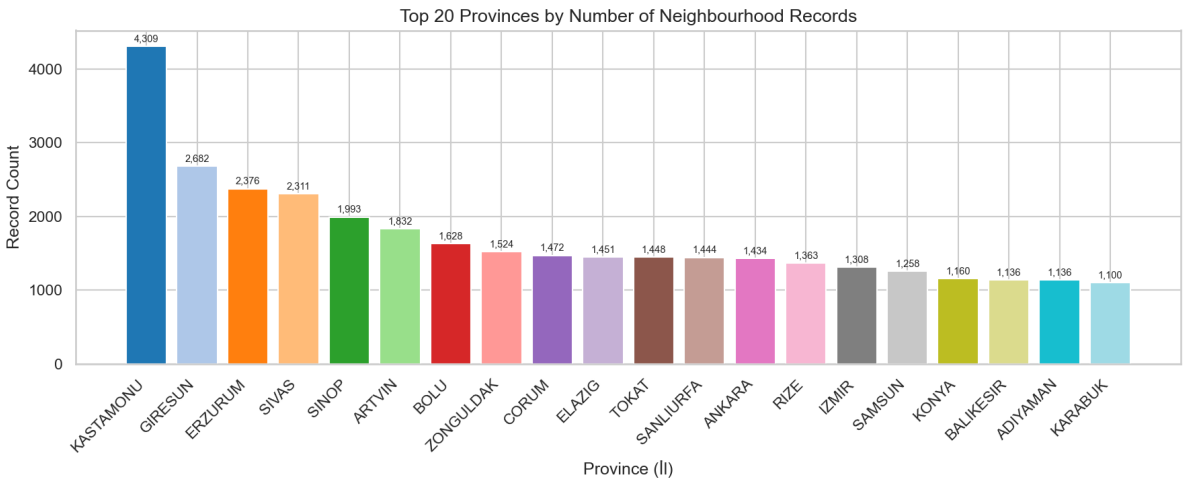


Figure 5: Top 20 provinces by number of records

Figure 5 reveals a counter-intuitive result: **Kastamonu, Giresun, and Sivas** lead in record count, not Istanbul or Ankara. This is because Black Sea and eastern Anatolian provinces contain many small rural villages, each registered as a separate neighbourhood entry. Urban provinces with fewer but larger neighbourhoods appear lower in the ranking.

6.4 Neighbourhoods per Postal Code

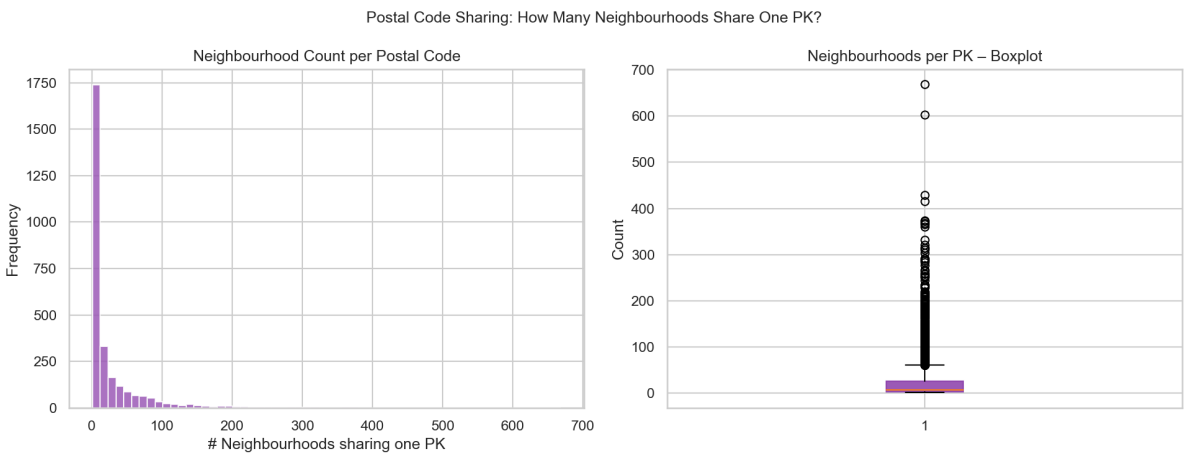


Figure 6: Distribution of neighbourhood count per postal code

The maximum number of neighbourhoods sharing a single postal code is **669** (PK 8890). The distribution is heavily right-skewed: most postal codes cover only a handful of neighbourhoods, but some district-level codes cover hundreds. This confirms that postal codes are assigned at the township or district level, not per-neighbourhood.

7 Relationships Between Features

7.1 Geographic Distribution

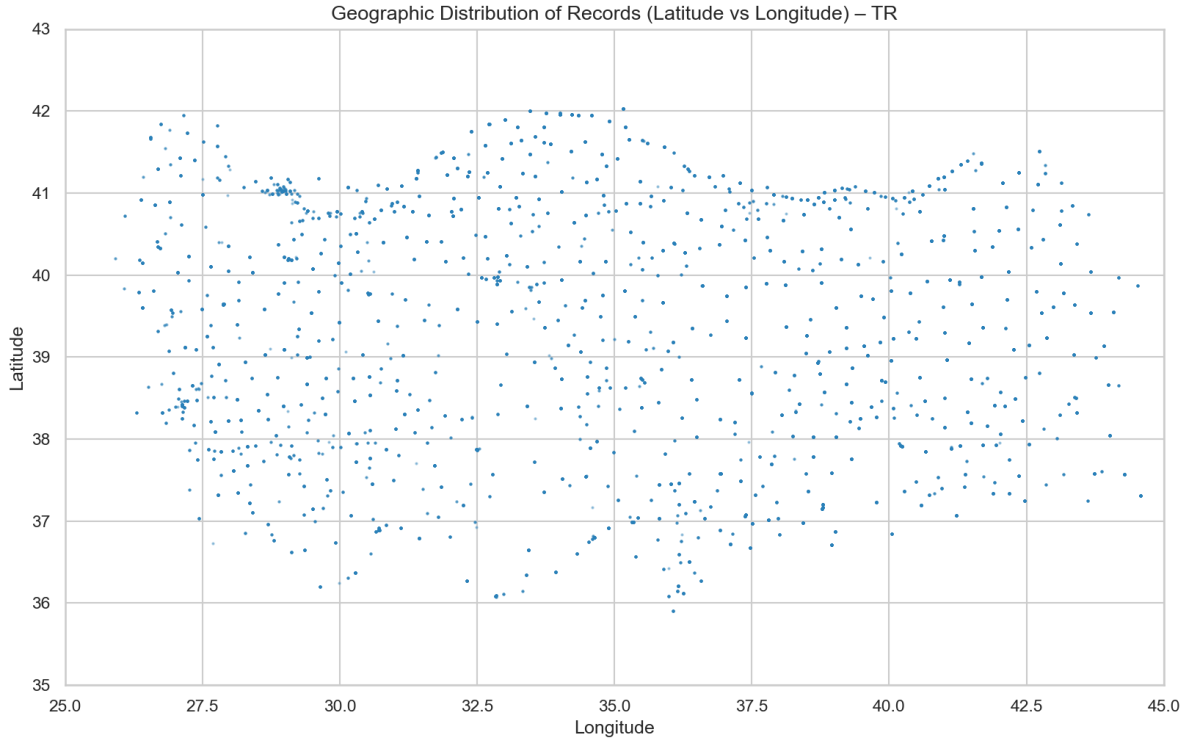
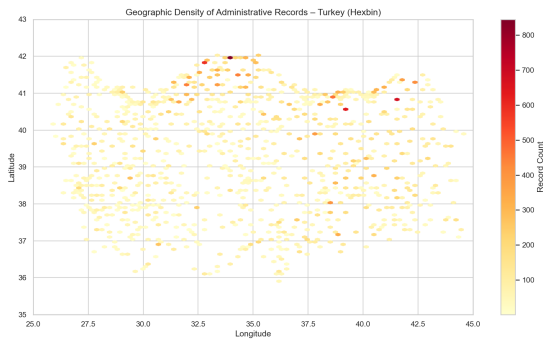
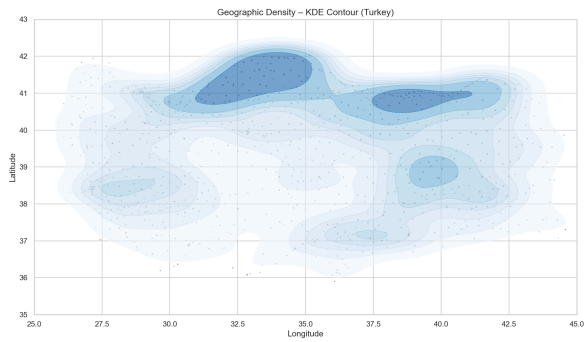


Figure 7: Scatter plot of latitude vs longitude (TR records)



(a) Hexbin density map



(b) KDE contour map

Figure 8: Geographic density of administrative records across Turkey

Figures 7 and 8 together trace the shape of Turkey remarkably well, confirming no outlier coordinates exist. The density maps reveal that administrative record density is highest in the **north-central and eastern regions**, corresponding to provinces with many rural villages. The Aegean and Marmara coasts, despite having large populations, show lower administrative record density due to more consolidated urban neighbourhoods.

7.2 Correlation Analysis

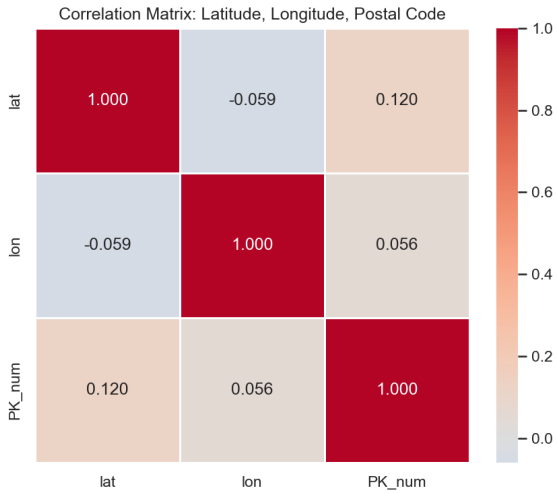


Figure 9: Correlation matrix: Latitude, Longitude, and Postal Code (numeric)

As shown in Figure 9, all pairwise correlations are weak:

- **Latitude vs Longitude:** $r = -0.059$ — near zero, as expected given Turkey's irregular shape.
- **Latitude vs PK:** $r = 0.120$ — very weak positive; postal codes loosely increase from south (lower plate numbers) to north.
- **Longitude vs PK:** $r = 0.056$ — near zero; postal codes are assigned administratively (by plate number), not geographically from west to east.

7.3 Latitude Distribution by Province

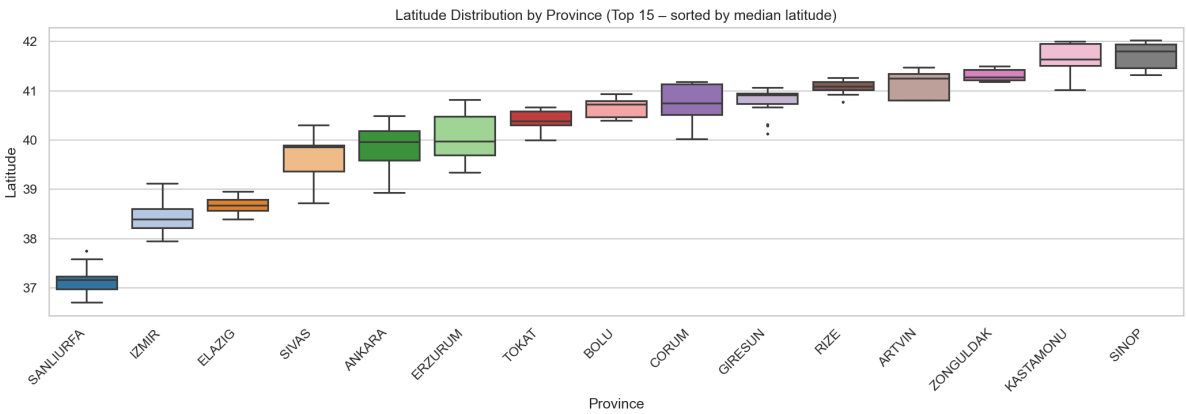


Figure 10: Latitude distribution per province (top 15, sorted by median latitude)

Figure 10 shows the expected north–south ordering of provinces when sorted by median latitude. Provinces with wider interquartile ranges (e.g. Kastamonu, Sinop) have administrative units spread across a larger north–south extent.

8 Anomaly and Error Detection

8.1 Postal Code Range Check

All Turkish postal codes fall within the expected range 01000–81999. All KKTC codes are within 99000–99999. No out-of-range postal codes were found.

8.2 Coordinate Bounding Box Check

All coordinates fall within Turkey's geographic bounding box (Latitude: 35°N–43°N, Longitude: 25°E–45°E). No erroneous coordinates were detected.

8.3 District Coordinate Granularity

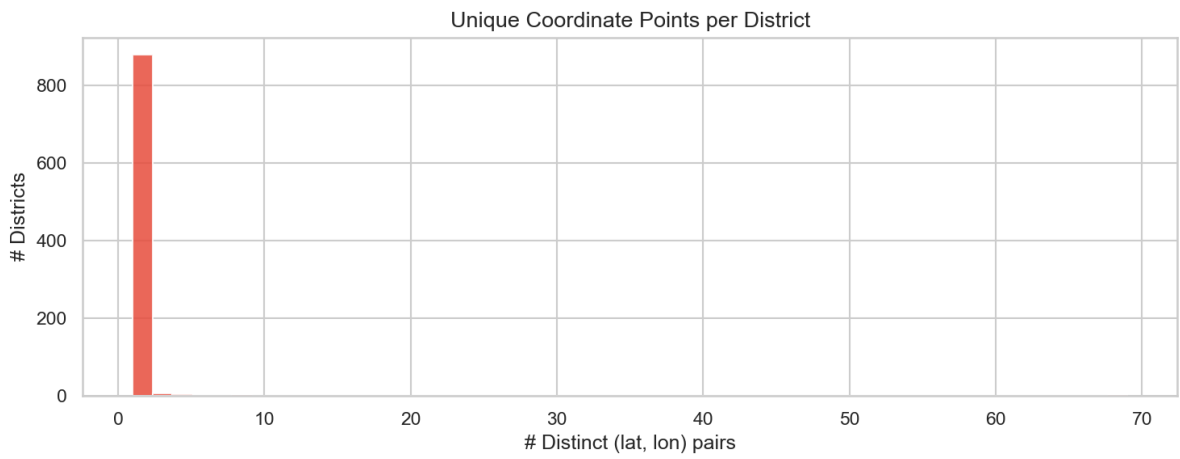


Figure 11: Number of unique coordinate points per district

Figure 11 shows a striking data quality issue: **833 out of 893 TR districts** have all their neighbourhood records assigned the *same single coordinate point*. Only 60 districts contain multiple distinct coordinates. This confirms that the dataset stores **district-level centroids** rather than precise per-neighbourhood GPS coordinates. Consequently, 79.4% of records share a coordinate with every other record in their district.

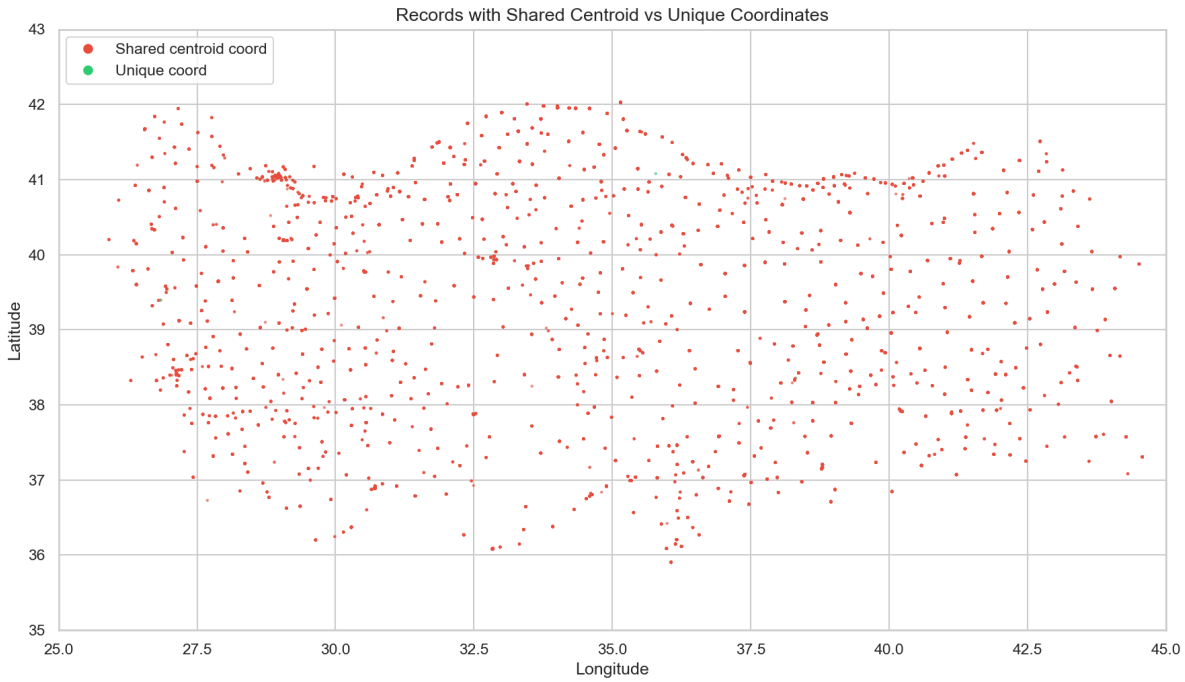


Figure 12: Records with shared centroid (red) vs unique coordinates (green)

9 Postal Code – Province Validation

Turkish postal codes follow the format `PPNNN`, where `PP` is the province's official plate number (01 for Adana, 06 for Ankara, 34 for Istanbul, etc.) and `NNN` is a suffix. After zero-padding all PKs to 5 digits (some had lost their leading zero during storage), **100% of records pass the validation**: every postal code prefix matches the province it belongs to.

This also reveals an important preprocessing note: PKs must be treated as strings and zero-padded to 5 digits before any numeric analysis.

10 Postal Code Gap Analysis

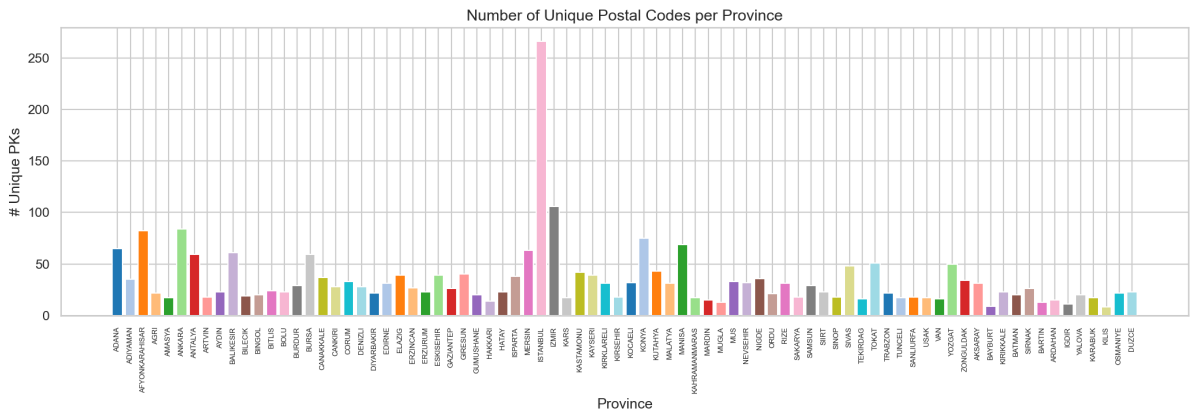


Figure 13: Number of unique postal codes per province

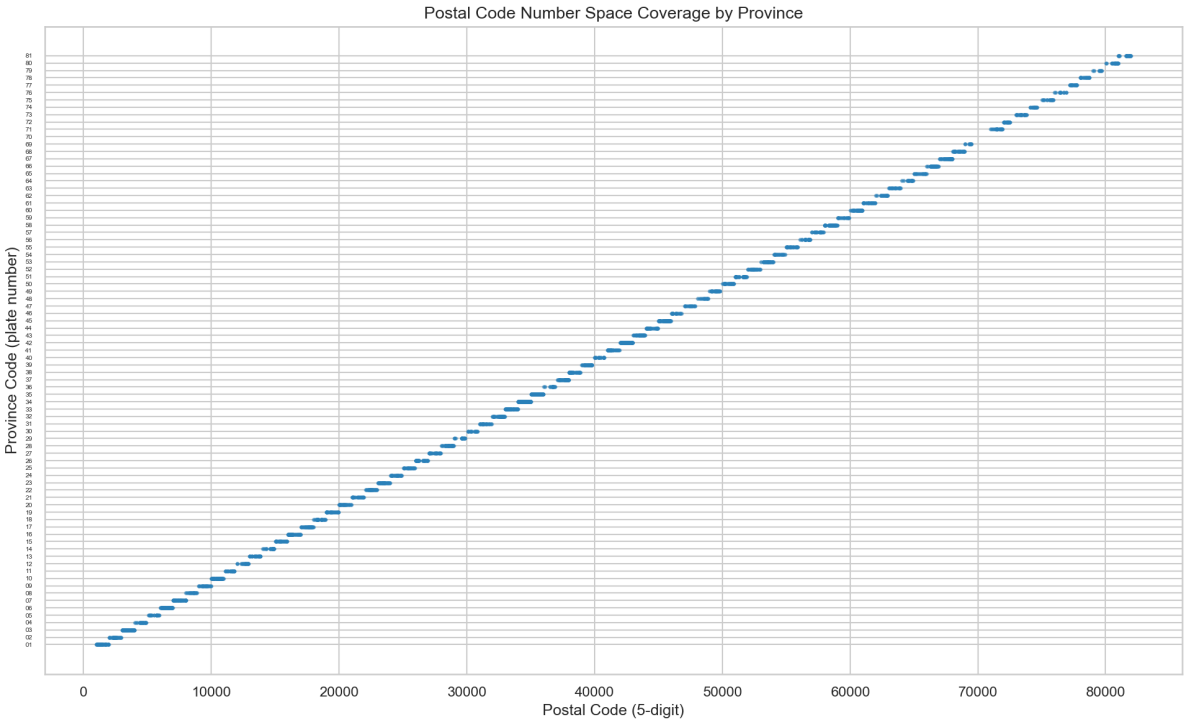
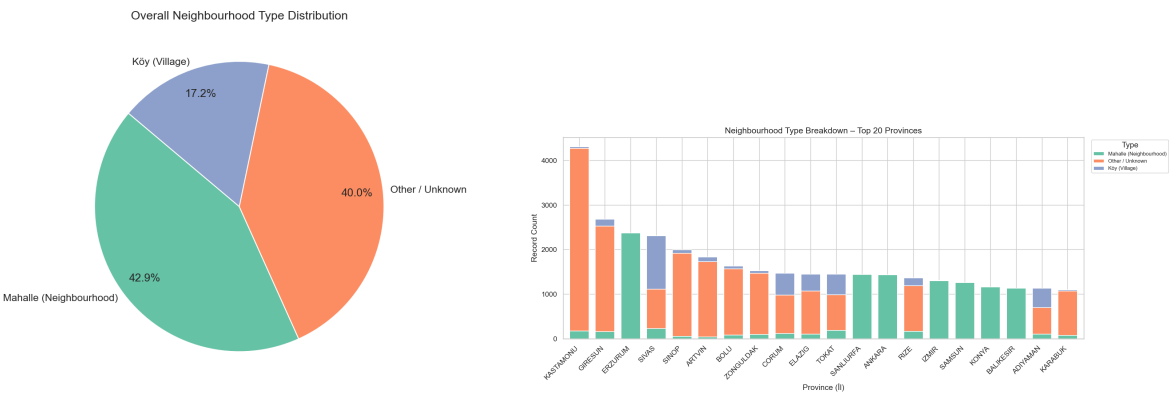


Figure 14: Postal code number space coverage by province

Figures 13 and 14 reveal that **province code 70 (Karaman) is entirely absent** from the dataset, despite being a valid Turkish province with plate number 70. All other 80 province codes (01–81, excluding 70) are present. This is a notable data gap that should be flagged.

Ankara (06) has the most unique postal codes (84), reflecting its large number of distinct urban districts. Provinces with few postal codes are typically smaller or less subdivided.

11 Neighbourhood Type Classification



(a) Overall type distribution (b) Type breakdown by province (top 20)

Figure 15: Neighbourhood administrative type classification

Neighbourhood names were classified by their administrative suffix:

- **Mahalle (MAH)**: 31,857 records (42.85%) — urban neighbourhoods
- **Köy/Köyü (KOY/KOYU)**: 12,769 records (17.18%) — villages
- **Other/Unknown**: 29,716 records (39.97%) — names with no standard suffix

The 40% "Other" category is itself a finding: a large portion of entries use non-standard naming, including parenthetical annotations (e.g., `CUHALAR MAH (GUNEYALUC KOYU)`) and KKTC-specific naming conventions.

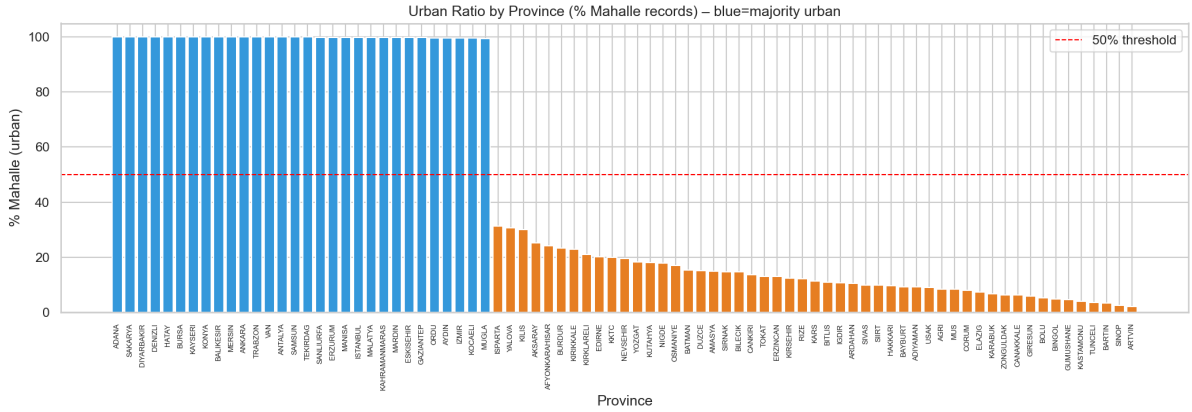


Figure 16: Urban ratio by province (% Mahalle records)

Figure 16 shows a clear divide: fully urbanised provinces (Adana, Istanbul, Ankara, Bursa, etc.) have 100% Mahalle records, while rural Black Sea and eastern provinces (Artvin 2.1%, Sinop 2.5%, Kastamonu 4.0%) are almost entirely composed of Köy entries.

12 Word Cloud of Neighbourhood Names

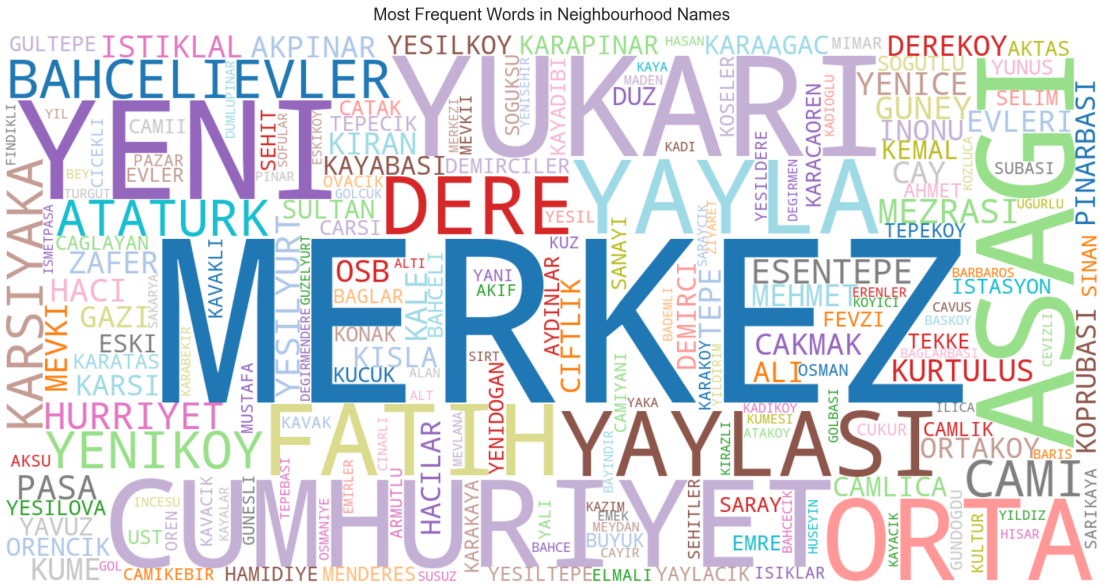


Figure 17: Word cloud of the most frequent content words in neighbourhood names

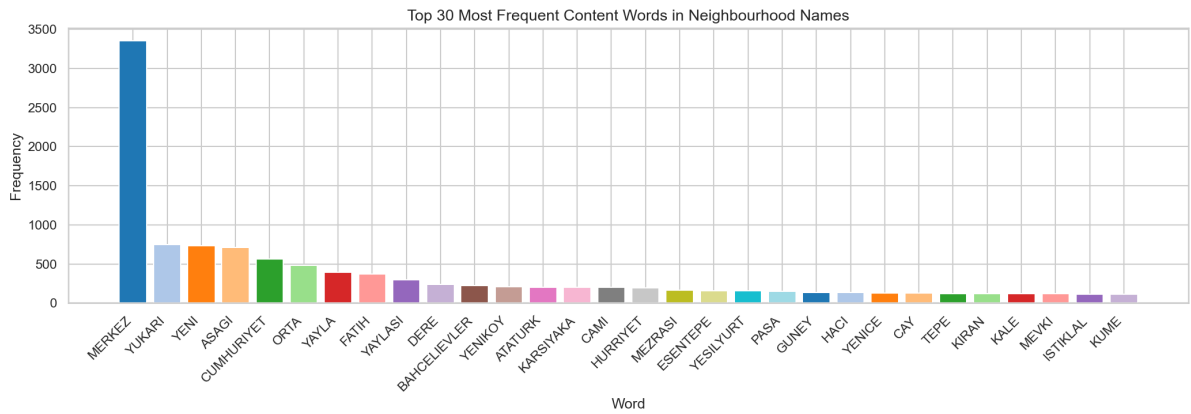


Figure 18: Top 30 most frequent content words in neighbourhood names

After removing administrative suffixes and cleaning punctuation, the most frequent words in neighbourhood names are **MERKEZ** (centre), **YUKARI** (upper), **YENI** (new), **ASAGI** (lower), and **CUMHURIYET** (republic). Directional words (Yukari/Asagi, Orta) are common because many villages are split into upper, lower, and middle parts. Names referencing Atatürk, Fatih, and Cumhuriyet reflect Turkey's republican political history embedded in place names.

13 Coordinate Precision Analysis

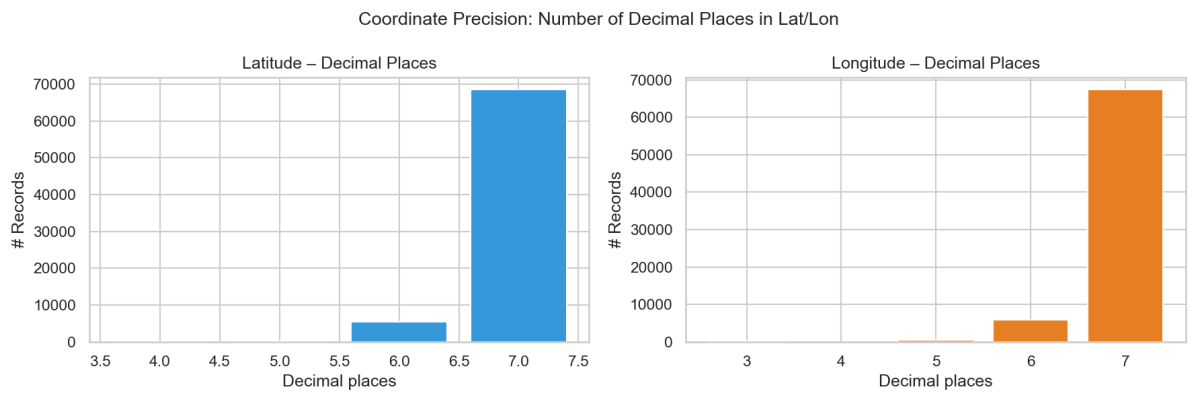


Figure 19: Distribution of decimal places in latitude and longitude values

Figure 19 shows that the majority of coordinate values are stored with **7 decimal places** (68,457 latitude values, 67,364 longitude values), which superficially suggests GPS-level precision (∼1 cm). However, this is misleading: as established in Section 8.3, 79.4% of records share an identical centroid with all other records in their district. The high decimal-place count reflects the format of the centroid value stored, not genuine per-neighbourhood GPS precision.

14 Key Findings and Conclusions

This section summarises the most important findings from the analysis.

14.1 Data Quality Issues

1. **KKTC records lack coordinates.** All 271 Northern Cyprus records have no latitude or longitude, making any spatial analysis of KKTC impossible with this dataset.
2. **Erzurum province is fully duplicated.** 2,376 of 2,378 duplicate records belong to Erzurum, indicating the province was entered twice during dataset construction. After deduplication, the true record count is 73,153.
3. **Coordinates are district-level centroids, not per-neighbourhood GPS.** 79.4% of all records (58,806 out of 74,071 TR records) share a single coordinate point with every other record in their district. Only 60 of 893 districts have multiple distinct coordinate values. The 7-decimal-place precision is therefore misleading.
4. **Karaman province (code 70) is missing.** Despite all 81 Turkish provinces being expected, no postal codes with the prefix `70` (Karaman) appear in the dataset.
5. **Leading zeros stripped from postal codes.** PKs stored as numbers lose their leading zero (e.g., 01720 becomes 1720). Zero-padding to 5 digits is required before any prefix-based analysis.

14.2 Structural and Geographic Insights

1. **Black Sea and eastern provinces dominate record counts**, not Istanbul or Ankara. Rural provinces with many small villages generate more administrative entries than densely populated urban provinces with larger but fewer neighbourhoods.
2. **Postal codes are coarse.** With only 2,818 unique codes for 74,342 records, a single code covers on average 26 neighbourhoods — and up to 669 in extreme cases. The postal system operates at the district level.
3. **40% of neighbourhood names use no standard suffix.** The classification into Mahalle/Köy/Belde covers only ∼60% of entries; the rest use non-standard or compound naming conventions.
4. **Place names reflect Turkey's history and geography.** The most frequent content words — Merkez, Yukari/Asagi, Yeni, Cumhuriyet, Atatürk, Fatih — reveal how Turkey's political history, topography, and religious heritage are embedded in its administrative geography.